

S.A. ENGINEERING COLLEGE

Autonomous Institution | Affiliated to Anna University | NBA & NAAC 'A' Grade

Dept of Electronics & Communication Engg & Dept of Electronics Engg [VLSI Design & Technology]

KERNEL PRIME '26 - SOFTWARE TRACK

National Level 24-Hour Hackathon | October 8-9, 2026

OFFICIAL PROBLEM STATEMENT: PS-05

Zero-Trust Cyber Threat Hunting & Autonomous Incident Response Copilot

Domain: Cybersecurity / LLM Agents / Graph Threat Modeling

1. PROBLEM OVERVIEW & CHALLENGE STATEMENT

Build an autonomous SOC security analyst engine using specialized LLMs and graph neural networks to reconstruct attack kill chains and deploy automated containment playbooks.

2. MANDATORY ARCHITECTURAL REQUIREMENTS

[REQ-1] Real-time ingestion and correlation of Zeek, Suricata, Sysmon, and CloudTrail logs.

[REQ-2] Automated MITRE ATT&CK technique mapping and attack progression graph construction.

[REQ-3] Interactive conversational analyst copilot with tool-use for automated firewall rule isolation.

[REQ-4] Zero-hallucination deterministic verification of security policies before execution.

[REQ-5] Automated synthetic malware sandbox detonation and behavioral signature generation.

3. HACKATHON EVALUATION MATRIX

* System Architecture & Concurrency Resilience: 25%

* Technical Innovation & Production Viability: 25%

* Live Demo & Real-World Latency Benchmarks: 30%

* Security Hardening, UI Polish & Documentation: 20%

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